

Personal Web Application.

A responsive, production-grade portfolio with live API integrations, dark/light theming, scroll-reveal animations, and accessibility built in.

Muhannad Al Melaifi

Software Engineering · 202253960

01

What I built

A single-page portfolio with many features in pure HTML, CSS, and JavaScript

3

Project cards

2

Live APIs

100%

Bug-free



Sticky nav

Active section highlight via Intersection Observer



Dark / Light

Theme persists in localStorage



Fully responsive

Mobile, tablet, desktop layouts



Project search

Live filter, level filter, sorting



Live APIs

GitHub stats + Zen quote with caching



Accessible

ARIA, focus styles, skip-link, semantic HTML

Technical deep dive

Clean structure, consistent design, honest about what's next.

STRUCTURE



Organized by concern

index.html for structure, css/styles.css for design, js/script.js for behavior. Images in assets/, docs in docs/. No build step needed.

CHALLENGE



Many distinct features

Combining theming, animations, search, APIs, and forms risked clutter. I split JavaScript into clear sections and reused the same card, button, and spacing patterns everywhere.

FUTURE WORK



Real backend form

The contact form validates on the frontend only the success toast is a UI simulation. Next step: wire it to a backend service or email API for real delivery.

AI as a co-pilot, not a replacement

Tools used

Claude (Anthropic)

Feature planning, code review, documentation drafts

GitHub Copilot

In-editor autocomplete for repetitive HTML and CSS

LESSONS LEARNED

AI supports — it doesn't replace.

AI was strongest as a thinking partner: drafting, suggesting, reviewing. The understanding still had to be mine — every snippet was tested in the browser, and I rewrote whatever didn't fit the project's style.



Every AI suggestion was reviewed, tested in the browser, and edited before it shipped.

Thank you.

From four assignments to a portfolio showing my academic journey



LIVE DEPLOYMENT

muhammad-melaifi.github.io/202253960-MuhammadAlMelaifi-assignment4



REPO

Muhammad-Melaifi